

and presents a greater thickness of armor for a projectile striking it horizontally to pass through.

Smoke

A means of hiding the movements of a vehicle or unit. Smoke can be dispensed by injecting fuel into a tank’s exhaust, activating a vehicle’s smoke grenade launchers, or firing a shell from a tank’s main gun. Modern smoke works in both the visible and infrared ends of the spectrum.

Smoothbore

A cannon designed to fire fin-stabilized rather than rotating projectiles, and so lacking interior rifling. Because they do not spin, the projectiles travel faster and so have greater armor penetration.

Spalling

Flakes broken off armor plate after the impact of a projectile. Some tanks have spall liners as a defense against high-velocity spalling.

Sponson

A gun platform projecting from the side of a tank.

Spotting gun

A small-caliber rifle or machine gun used as a ranging device for tank guns. They were used as an alternative to optical range finders until the development of the laser range finder.

Spring

The part of a suspension system that both absorbs the upward movement of the wheels when on rough terrain and keeps the wheels pressed onto the ground.

Sprocket

A cogged wheel that meshes with a tank’s track to give the track linear motion. Sprockets are usually the only powered wheels on an AFV.

Squadron

A military unit, normally equivalent in size to the company and consisting of around 150 soldiers or 14–18 tanks. “Squadron” was traditionally a cavalry term. In the US Army it is the equivalent of a battalion.

Stalemate

A tactical impasse on a battlefield. The stalemate between the Allied and German armies on the Somme during World War I was caused by both sides being dug in and defended by machine guns and artillery. The tank was developed by the British specifically to end this stalemate.

Strategy

The overall plan of a campaign. Strategic objectives determine the tactical deployment of troops and materiel.

Super-heavy tank

A tank of a size and mass greater than that of a heavy tank.

Tactics

The means by which particular military objectives are met, as opposed to strategy, which concerns the overall aim of a campaign.

Tandem warhead

A feature of recent ATGMs, intended to defeat ERA. The first warhead detonates and triggers the ERA, the second follows a short period later and is able to penetrate the vehicle’s armor, which no longer benefits from the ERA.

Tank

An AFV designed for front-line combat, featuring strong armor, heavy firepower, and tracks for battlefield maneuverability. Its name derives from the secrecy under which it was conceived—engineers were told that they were working on a new design of water tank.

Tank destroyer

An AFV armed with a direct-fire gun or missile launcher designed specifically to target enemy armored vehicles.

Tankette

A tracked AFV resembling a small tank, designed for scouting and light infantry support. Tankettes saw wide use during the interwar years and World War II, particularly in the Imperial Japanese Army, but have since ceased production because they were found to be too lightly armed and armored to survive on the battlefield.

Thermal sleeve

An insulating device that is placed around the barrel of a main gun. It ensures that the temperature of the barrel remains even: differences can cause the metal to expand, and so affect accuracy.

Titanium

A strong but relatively lightweight metal used in tank armor.

Top attack

A method used by modern ATGMs to overcome increasingly capable composite armor. The missile flies over the tank and detonates above it. This directs the warhead at the thinner roof armor.

Torsion bar

A suspension system that uses a twisting metal bar to cushion the vehicle’s movement.

Tracer

A bullet with a pyrotechnic charge in its base. The charge is ignited when the round is fired and shows its trajectory. Tracer helps gunners to direct their fire, especially in circumstances where sights would be less effective, such as in darkness.

Track

The continuous belt running through or around the geared sprocket wheel, idler, road wheels, and return rollers of a tank.

Transmission

The electrical, hydraulic, or mechanical means by which power from an engine is converted into the rotary motion of a vehicle’s wheels or tracks.

Traverse

The ability of a gun or turret to rotate from the centerline of its mount. A fully revolving gun or turret is said to have a traverse of 360 degrees.

Trench

The field fortification that the tank was designed to overcome. During World War I, strong networks of continuous trenches protected by machine-guns and artillery created a stalemate on the Western Front, and only tanks proved capable of breaking it.

Trim vane

A hinged metal screen that can be extended before a vehicle enters a body of water. This reduces the risk of it being swamped by large amounts of water washing over the front.

Troop

A military unit, normally equivalent in size to the platoon and consisting of around 30 soldiers or 3–5 tanks. Troop was traditionally a cavalry term. In the US Army it is the equivalent of a company.

Turret

The rotating top section of a tank, accommodating the main gun and most of the crew, usually the commander, gunner, and loader. The first turreted tank was the Renault FT of 1917.

V-shaped hull

A design feature that angles the underside of a vehicle upward. When viewed from the front or rear the lower hull is shaped like a V. This deflects a mine blast outwards away from the vehicle, rather than upwards into the crew compartment.

V-twin engine

An engine design with two banks of cylinders arranged in a “V” formation.

Volute-spring suspension

A type of tank suspension featuring a compression spring shaped like a cone, or volute, mounted in a road-wheel bogie for a pair of wheels. Commonly used in US and Italian tanks during World War II, it proved more effective than contemporary spring, leaf-spring, or torsion-bar suspension systems.

Warhead

The part of a projectile that contains the explosive. Other parts can include a guidance system or a fuse.

Warsaw Pact

A defense treaty between the Soviet Union and the Soviet satellite states of Bulgaria, Czechoslovakia, East Germany, Hungary, Poland, Romania, and Albania. Signed in 1955, the treaty established a counterweight to NATO.

Wedge

A formation of tanks arranged in a triangular shape.